

Problem Set 8

Problem 1: Exercise 7.3.1

Plug the power series expansion $\psi = \sum_{n=0}^{\infty} c_n y^n$ into the equation $\psi'' + (2\varepsilon - y^2)\psi = 0$ to get

$$\sum_{n=0}^{\infty} c_n [n(n-1)y^{n-2} + (2\varepsilon - y^2)y^n] = 0.$$

Shift $n \rightarrow n+2$ in the first term, and $n \rightarrow n-2$ in the third term to get

$$\sum_{n=0}^{\infty} y^n [(n+2)(n+1)c_{n+2} + 2\varepsilon c_n - c_{n-2}] = 0$$

with the convention that

$$c_{-2} = c_{-1} = 0.$$

This implies

$$c_{n+2} = \frac{c_{n-2}}{(n+1)(n+2)} - \frac{2\varepsilon c_n}{(n+1)(n+2)}$$

for all $n \geq 0$.

Problem 3: Exercise 7.3.7

In the momentum basis $|\psi\rangle \rightarrow \psi(p)$, $P \rightarrow p$, and $X \rightarrow i\hbar(d/dp)$, so the energy eigenvalue equation

$$\left(\frac{1}{2m} P^2 + \frac{m\omega^2}{2} X^2 \right) |E\rangle = E|E\rangle$$

becomes

$$\frac{1}{2m} p^2 \psi(p) - \frac{m\omega^2}{2} \hbar^2 \psi''(p) = E \psi(p).$$

Compare this to the position-basis equation

$$\frac{m\omega^2}{2} x^2 \psi(x) - \frac{1}{2m} \hbar^2 \psi''(x) = E \psi(x).$$

These are the same equations with the substitutions $x \leftrightarrow p$ and $m \leftrightarrow 1/(m\omega^2)$.

Problem 2: Exercise 7.3.5

$$\begin{aligned}\langle n|X|n\rangle &= \int_{-\infty}^{\infty} dx \psi_n^*(x) \cdot x \cdot \psi_n(x) = \int_{-\infty}^{\infty} dx x \cdot \psi_n^2(x) \quad \text{since } \psi_n(x) \text{ is real} \\ &= 0 \quad \text{since } x \text{ is odd and } \psi_n^2(x) \text{ is even.}\end{aligned}$$

$$\begin{aligned}\langle n|P|n\rangle &= \int_{-\infty}^{\infty} dx \psi_n^*(x) (-i\hbar) \frac{d}{dx} \psi_n(x) = (-i\hbar) \int_{-\infty}^{\infty} dx \psi_n \psi_n' = -\frac{i\hbar}{2} \int_{-\infty}^{\infty} dx (\psi_n^2)' \\ &= -\frac{i\hbar}{2} \psi_n^2 \Big|_{-\infty}^{\infty} = 0 \quad \text{since } \psi_n \rightarrow 0 \text{ as } |x| \rightarrow \infty.\end{aligned}$$

$$\begin{aligned}\langle 1|X^2|1\rangle &= \int_{-\infty}^{\infty} dx \psi_1^* x^2 \psi_1 = \left(\frac{m\omega}{4\pi\hbar}\right)^{1/2} \int_{-\infty}^{\infty} dx x^2 \left[2x \left(\frac{m\omega}{\hbar}\right)^{1/2}\right]^2 e^{-m\omega x^2/\hbar} \\ &= \frac{2}{\sqrt{\pi}} \left(\frac{m\omega}{\hbar}\right)^{3/2} \int_{-\infty}^{\infty} dx x^4 e^{-m\omega x^2/\hbar} = \frac{2}{\sqrt{\pi}} \left(\frac{m\omega}{\hbar}\right)^{3/2} \frac{3\sqrt{\pi}}{4} \left(\frac{m\omega}{\hbar}\right)^{-5/2} \\ &= \frac{3\hbar}{2m\omega}.\end{aligned}$$

$$\begin{aligned}\langle 1|P^2|1\rangle &= \int_{-\infty}^{\infty} dx \psi_1^* (-i\hbar)^2 \psi_1'' = -\hbar^2 \frac{2}{\sqrt{\pi}} \left(\frac{m\omega}{\hbar}\right)^{3/2} \int_{-\infty}^{\infty} dx x e^{-m\omega x^2/2\hbar} \left(x e^{-m\omega x^2/2\hbar}\right)'' \\ &= -\hbar^2 \frac{2}{\sqrt{\pi}} \left(\frac{m\omega}{\hbar}\right)^{3/2} \int_{-\infty}^{\infty} dx x^2 \left(\frac{m\omega}{\hbar}\right) \left[\left(\frac{m\omega}{\hbar}\right) x^2 - 3\right] e^{-m\omega x^2/\hbar} \\ &= -\hbar^2 \frac{2}{\sqrt{\pi}} \left(\frac{m\omega}{\hbar}\right)^{5/2} \cdot (-1) \frac{3\sqrt{\pi}}{4} \left(\frac{m\omega}{\hbar}\right)^{-3/2} = \frac{3m\omega\hbar}{2}.\end{aligned}$$

$$\Delta X^2 = \langle 0|X^2|0\rangle = \sqrt{\frac{m\omega}{\pi\hbar}} \int_{-\infty}^{\infty} dx x^2 e^{-m\omega x^2/\hbar} = \left(\frac{m\omega}{\hbar}\right)^{1/2} \frac{\sqrt{\pi}}{2} \left(\frac{m\omega}{\hbar}\right)^{-3/2} = \frac{\hbar}{2m\omega}.$$

$$\begin{aligned}\Delta P^2 = \langle 0|P^2|0\rangle &= -\hbar^2 \sqrt{\frac{m\omega}{\pi\hbar}} \int_{-\infty}^{\infty} dx e^{-m\omega x^2/2\hbar} \left(e^{-m\omega x^2/2\hbar}\right)'' \\ &= -\frac{\hbar^2}{\sqrt{\pi}} \left(\frac{m\omega}{\hbar}\right)^{3/2} \int_{-\infty}^{\infty} dx \left(\frac{m\omega}{\hbar} x^2 - 1\right) e^{-m\omega x^2/\hbar} \\ &= -\frac{\hbar^2}{\sqrt{\pi}} \left(\frac{m\omega}{\hbar}\right)^{3/2} \cdot (-1) \frac{\sqrt{\pi}}{2} \left(\frac{m\omega}{\hbar}\right)^{-1/2} = \frac{m\omega\hbar}{2}.\end{aligned}$$

$$\therefore \Delta X \Delta P = \sqrt{\frac{\hbar}{2m\omega}} \cdot \sqrt{\frac{m\omega\hbar}{2}} = \frac{\hbar}{2}.$$